Alexandru Asandei

Computer Science Student | Software Engineer

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EDUCATION

Technical University of Cluj-Napoca (UTCN) — Faculty of Automation and Computer Science, Cluj-Napoca

B.S. in Computer Science (English Curriculum) | *Expected: 2028 (Currently in Year 3 of 4)*

Relevant Coursework: Data Structures & Algorithms, OOP (Java/C++), Operating Systems, Computer Architecture, Computer Networks, Database Systems, Software Engineering.

TECHNICAL SKILLS

Languages: C++, Java, Python, Dart, SQL, TypeScript, C

Frameworks & Libraries: Spring Boot (Java 21), FastAPI, Flutter, React, Node.js

Systems & Infrastructure: Multithreading, Win32 API (SendInput), Winsock2 (TCP/UDP Sockets), Docker & Compose, GCP Cloud Run, PostgreSQL (Neon), CI/CD (GitHub Actions)

Developer Tools: Linux / Bash, Git, CMake, Gradle, Postman, Visual Studio, IntelliJ IDEA, Android Studio

TECHNICAL PROJECTS

PhoneAsPeri — Modular Hardware Remote Control System | *C++, Flutter/Dart, Win32 API, Winsock2 UDP*
github.com/original4to4/PhoneAsPeri

- Engineered a distributed real-time remote control system connecting a Flutter mobile app to a native C++ Windows daemon over local Wi-Fi with **sub-10ms response latency** and a **<5 MB RAM footprint** (zero dynamic allocations in the UDP reception hot path).
- Designed one-time session authentication via dynamic QR codes with instant packet-level token verification on raw UDP sockets to discard unauthorized commands at the network boundary.
- Simulated hardware mouse and keyboard events via native Win32 `SendInput` API, handling relative cursor movement, dynamic wheel scrolling, and continuous multi-key press states for PC gaming (WASD).
- Developed an interactive multi-touch Flutter studio with 64-key virtual PC keyboard, 8-directional analog joystick, and a Gyroscope **"Air Mouse"** with clutch control and noise deadzone filtering.

LDD_FRS — Full-Stack AI Football Analytics Platform | *Python, Java 21, Spring Boot, FastAPI, GCP, React*
"U" Hackathon Team Project | github.com/TartaTudorAndrei/LDD_FRS

- Co-developed a predictive sports analytics platform in a 4-person team during the **"U" Hackathon**, orchestrating microservices across Java 21, Python, and React.
- Architected the **Player Performance Index (PPI)** microservice in Python (FastAPI), engineering position-weighted ranking algorithms and mathematical normalization models to evaluate player performance trends.
- Implemented automated database persistence with **Neon Serverless PostgreSQL** and containerized multi-module microservices using **Docker Compose** for Google Cloud Platform (GCP Cloud Run) deployment.

Rentz Multiplayer Card Game — Desktop Application | *Java, Swing, TCP Sockets, PostgreSQL*
github.com/original4to4/LabProject_RENTZ_game

- Designed an object-oriented desktop card game engine in Java enforcing full Rentz tournament rules, turn validation, and state machines with a Java Swing GUI.
- Developed a client-server multiplayer network layer using **Java TCP sockets** for real-time game synchronization and persistent user scoring backed by PostgreSQL.

Erudit — Gamified Educational Mobile App | *Flutter / Dart, C++*
DPIT (Descoperă-ți pasiunea în IT) Program | github.com/aniteicristi/erudit

- Collaborated in a 6-person team during the DPIT mentorship program to build a Duolingo-style STEM learning mobile app (Flutter/Dart & C++) with interactive quizzes and streak mechanics.

HACKATHONS & EXTRACURRICULARS

- **"U" Hackathon Cluj-Napoca** — Team Developer (LDD_FRS Full-Stack AI Football Analytics Platform; Java 21 Spring Boot, Python FastAPI, GCP, React)
- **Descoperă-ți pasiunea în IT (DPIT)** — 6-Month Practical Mentorship Academy & Team Competition in Cluj-Napoca (Erudit Mobile Learning App)